



Molecular Biology

Genome-wide expression analysis of rice ABC transporter family across spatio-temporal samples and in response to abiotic stresses



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ABSTRACT

Although the super family of ATP-binding cassette (ABC) proteins plays key roles in the physiology and development of plants, the functions of members of this interesting family mostly remain to be clarified, especially in crop plants. Thus, systematic analysis of this family in rice (*Oryza sativa*), a major model crop plant, will be helpful in the design of effective strategies for functional analysis. Phylogenomic analysis that integrates anatomy and stress meta-profiling data based on a large collection of rice Affymetrix array data into the phylogenetic context provides useful clues into the functions for each of the ABC transporter family members in rice. Using anatomy data, we identified 17 root-preferred and 16-shoot preferred genes at the vegetative stage, and 3 pollen, 2 embryo, 2 ovary, 2 endosperm, and 1 anther-preferred gene at the reproductive stage. The stress data revealed significant up-regulation or down-regulation of 47 genes under heavy metal treatment, 16 genes under nutrient deficient conditions, and 51 genes under abiotic stress conditions. Of these, we confirmed the differential expression patterns of 14 genes in root samples exposed to drought stress using quantitative real-time PCR. Network analysis using RiceNet suggests a functional gene network involving nine rice ABC transporters that are differentially regulated by drought stress in root, further enhancing the prediction of biological function. Our analysis provides a molecular basis for the study of diverse biological phenomena mediated by the ABC family in rice and will contribute to the enhancement of crop yield and stress tolerance.

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Introduction

ATP-binding cassette transporter family (ABC-transporter) constitutes one of the largest protein families with transporter activity and is conserved in all organisms. This family plays roles in the export or import of various substrates across biological membranes. Minimal structure of ABC transporter family requires highly conserved nucleotide-binding domains (NBD) and a less conserved trans-membrane domain (TMD). The former consists of Walker A and Walker B motifs for ATP binding and H- and Q-loops for the hydrolysis. The latter forms the ligand binding sites by several α -helices and determines the specificity of substrates. A typical membrane-bound ABC transporter contains four domains: two TMDs and two NBDs. ABC transporter is formed from single

gene or two genes (heterodimers or homodimers) (Kang et al., 2011).

By the difference in protein size, orientation (the organization of domain), and TMDs sequence (idiotypic and/or linker), plant ABC transporter family is currently classified to eight subfamilies. From ABCA to ABCD, subfamilies are characterized in forward TMD–NBD organization, whereas ABCG subfamily contains reverse orientation NBD–TMD. The soluble subfamilies E and F only encode two NBDs but no TMD, and the members of ABCI subfamily retain single domain such as NBD or accessory domain. Even TMD sequences were selected as criterion for the subfamily classification except soluble ABC transporters, high analogy among their sequences do not entirely indicate similarity in their function. However, a few closely linked members in their phylogenetic tree are more likely to play similar or redundant functions. For example, *AtABCB1* and its closest homolog *AtABCB19*, participate in auxin transport (Sidler, 1998), or *AtABCC1* and *AtABCC2* exhibit redundant function in the translocation of plant vacuolar phytochelatin (Song et al., 2010). Beyond the range of species, functions of genes with homology are frequently conserved. For example, Qin et al. (2013) provided evidence that *OsABCG15* is essential for post-meiotic anther and pollen exine development, like the homolog of *OsABCG15* in Arabidopsis,

Abbreviations: ABA, abscisic acid; ABC transporter, ATP-binding cassette transporter; As, arsenate; Cd, cadmium; COV, coefficient of variation; Cr, chromium; NBD, nucleotide binding domain; Pb, lead; RT-PCR, reverse transcription polymerase chain reaction; TMD, trans-membrane domain.

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AtABCG26/WBC27 (Quilichini et al., 2010). Another example is *AtABCG32/PEC1* and *OsABCG31* that are involved in the formation of the cuticular layer of the cell wall to protect from losing water (Bessire et al., 2011; Chen et al., 2011). Therefore, phylogenetic analysis among different species provides us useful clues for the functional study among homologous genes. Even if the structural feature has been well studied, the identification of biological functions in plants is very limited. In case of *Arabidopsis* ABC transporters, there are 128 family members and functions of 26 ABC transporters have been characterized (Kang et al., 2011), and the number for rice ABC transporters with known functions is much smaller than that of *Arabidopsis*. Until now, three ABC transporters related to water retention in leaf, pollen exine development, and production of phytic acid in rice have been functionally characterized (Chen et al., 2011; Qin et al., 2013; Xu et al., 2009).

Water scarcity causes a huge damage on crop production. Besides their developmental roles, ABC transporters perform the function in rescuing plants which are exposed to various abiotic stresses, especially drought. So far, the identification of both the importer (Kang et al., 2010) and exporter (Kuromori et al., 2010) of ABA in *Arabidopsis* enhanced our knowledge about ABA signaling which is one of the most important phytohormone pathways associated with drought stress responses. ABC transporters also have functions related to guard cell responses (Klein et al., 2004; Kuromori et al., 2011) or required for functional cuticle formation, associated with water retention in the leaf like *AtABCG32/PEC1* and *OsABCG31* as mentioned earlier. Genome-wide analysis of ABC transporters in rice associated with water scarcity has not been carried out and will be required to provide novel insight of ABC transporter for rescuing water scarcity.

Completion of the rice genome sequence has revealed a large number of ABC transporters (IRGSP, 2005; Verrier et al., 2008). The main obstacle to their functional identification is the functional redundancy within this supergene family and the lack of information guiding functional analysis. Thus, a systematic overview of the ABC family in rice might be a useful starting point for further functional analyses under diverse stress conditions as well as for unique developmental process. Phylogenomics is a phylogenetic approach to predict the biological functions of members of large gene families by assessing the similarity among gene products (Cao et al., 2012; Jung et al., 2010a). The power of estimating functional redundancy is even more enhanced by combining phylogenetic analysis with a large amount of transcriptome data, such as integrating analysis of gene expression data related to SUB1A-mediated submergence tolerance with the phylogenetic tree of the AP2 family (Jung et al., 2010b). Recently, we demonstrated that selection of the predominant gene family member in light conditions has better performance in identifying defective phenotypes through the related loss-of-function study than selecting non-predominant family members (Jung et al., 2008).

In this study, we performed phylogenomic analysis of ABC transporter family members to provide functional clues of individual family members. The anatomy or stress meta-expression profiles based on a large collection of Affymetrix data were then integrated into the phylogenetic tree context of the ABC transporter family. As a result, we identified 38 genes that showed tissues/organ preferred expression, 3 genes that were ubiquitously expressed, 47 genes that responded to heavy metal, 16 genes associated with nutrient deficiency, and 51 genes associated with abiotic stress. Of them, expression patterns of 14 genes in response to drought stress in root are confirmed by quantitative RT-PCR analysis. We developed a functional gene network associated with nine ABC transporter genes differentially expressed in root after drought treatment and this network suggests a hypothetical model to elucidate the underlying molecular mechanisms, which might in turn facilitate future agricultural applications in rice for enhancing water use efficiency.

Materials and methods

Multiple sequence alignment and phylogenetic tree building

To perform phylogenomic analysis of ABC transporters in rice and *Arabidopsis thaliana*, we collected 125 family members with locus IDs from the Rice Genome Annotation Project (RGAP, <http://rice.plantbiology.msu.edu/>) (Table S1) and 128 *Arabidopsis* family members from a previous report (Verrier et al., 2008). We selected a representative transcript encoding the ABC protein for each locus in rice as suggested in RGAP. Multiple amino acid sequences were aligned using ClustalX program version 2.0.11 (Higgins et al., 1996). The phylogenetic analysis was performed by using MEGA 5.2 and under the following parameters: neighbor-joining tree method, complete deletion, and bootstrap with 500 replicates (Tamura et al., 2011). We developed a phylogenetic tree with rice and *Arabidopsis* ABC transporters. Rice ABC proteins are currently classified into eight subfamilies (Figs. 1 and S1) in accordance with new nomenclature, based on RGAP annotation release 7 (http://rice.plantbiology.msu.edu/downloads_gad.shtml). We also generated a phylogenetic tree for each subfamily except for ABCD and ABCE, which have only 2 and 3 members, respectively (Fig. S1).

Supplementary Table S1 and Fig. S1 related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2014.05.006>.

Meta-analysis of tissue specific expression profiles and stress responsive expression analysis

Integration of transcriptomes into the context of the phylogenetic tree might be a good reference to direct useful experimental strategies for further functional analysis. Therefore, we used meta-analysis of tissue specific expression profiles based on 983 Affymetrix array data downloaded from the NCBI gene expression omnibus (GEO, <http://www.ncbi.nlm.nih.gov/geo/>) (Cao et al., 2012). We then uploaded the log₂ normalized intensity data in tab-delimited text format into Multi Experiment Viewer (MEV, <http://www.tm4.org/mev/>) and illustrated them by using heat maps (Figs. 2–4 and S2 and Table S4). We separated the tissue samples into two groups according to development stage: vegetative stage and reproductive stage. We then selected genes that have at least 2-fold higher expression level in each tissue type relative to the other tissues with a *P*-value less than 0.05. In addition, we generated a heat map image based on log₂ fold-change data in response to abiotic stresses and then integrated the heat map into the phylogenetic tree context. We selected genes showing up-regulation under diverse stresses with log₂ changes greater than 1.5-fold or lower –1.5-fold for down-regulation with a coefficient of variation (COV) less than 1, compared with normal conditions (Fig. 5, Tables S2–3).

Supplementary Tables S2–S4 and Fig. S2 related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2014.05.006>.

Promoter analysis

To analyze the promoter regions for four loci (*OsABCG30*, *OsABCG41*, *OsABCG47*, and *OsABCG48*) differentially expressed between *japonica* and *indica*, we extracted 2 kb upstream sequences from start codon of four loci in both *japonica* and *indica* genomes from Gramene (http://plants.ensembl.org/Oryza_sativa/Info/Index) and EMBL (http://ensembl.griemene.org/Oryza_indica/Info/Index). Comparison of transcription factor binding sites (TFBSs) was processed by Plant promoter analysis navigator, Plantpan (<http://plantpan.mbc.nctu.edu.tw/>) (Chang et al., 2008).

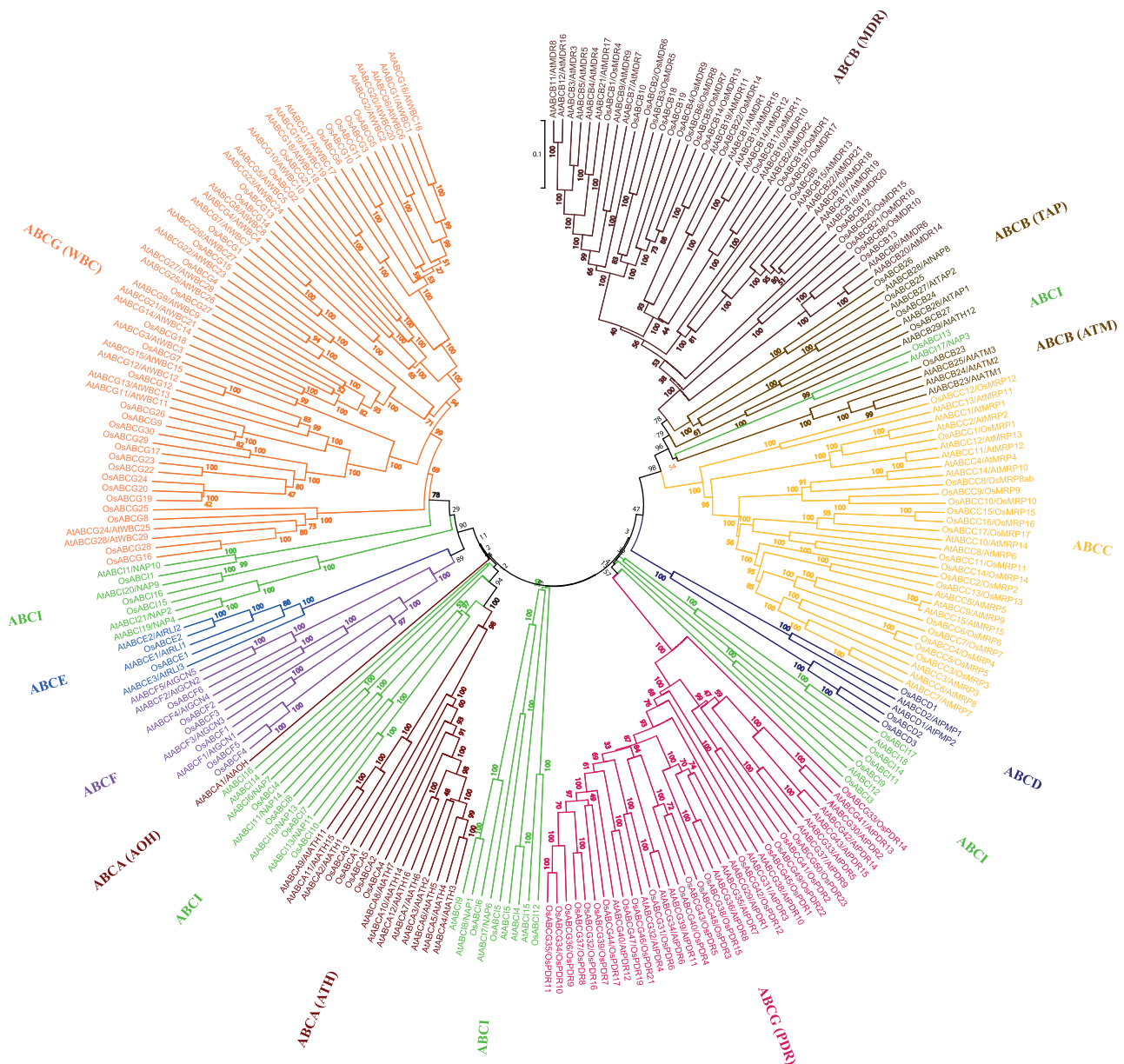


Fig. 1. Phylogenetic tree showing predicted relationships between the ABC family in rice and *Arabidopsis thaliana*. We classified 125 rice ABC transporter proteins and 128 *Arabidopsis* ABC transporter proteins into eight subfamilies. Each subfamily is marked by a different color and is consistent with the subgroup name. Numeric values indicate estimated evolutionary distances. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Fig. S3 shows the results of compared TFBSs and similarity alignment between *japonica* and *indica* promoter regions for 4 loci. Yellow colored regions indicate high homologs between compared *japonica* and *indica* sequences and the rest was not indicated by colored lines or regions.

Supplementary Fig. S3 related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2014.05.006>.

Plant material and drought treatment

Rice (*Oryza sativa*) seeds (Dongjin and IR64 varieties) were germinated on Murashige Skoog medium under controlled conditions of 28 °C day/25 °C night temperatures, 8-h light/16-h dark cycle, and 78% relative humidity after sterilization with 50% (w/v) commercial bleach for 30 min with gentle shaking. After 7 days of germination, the seedlings were washed with sterilized water to

discard agar and incubated in water for a day, then completely removed water and air-dried at 28 °C. Roots were harvested at 2 and 4 h after exposure to stress treatment, frozen in liquid nitrogen, and stored at -70 °C for further analysis (Jeong et al., 2010; Oh et al., 2009).

RNA extraction, semi-quantitative RT-PCR, and real-time PCR

Roots and shoots of rice seedlings were frozen in liquid nitrogen and ground with a Tissue Lyser II (Qiagen; Hilden, Germany). RNAs were extracted with the RNAiso Plus Kit according to the manufacturer's protocol (Takara Bio; Kyoto, Japan). For tissue specific expression patterns by RT-PCR, we used two control primer sets for rice ubiquitin 5 (*OsUbi5*, LOC.Os01g22490) and rice elongation factor-1 α (*OseEF-1 α* , LOC.Os03g08020) (Jain et al., 2006). The PCR cycle conditions used were 95 °C for 30 s, 57 °C for 30 s, and 72 °C

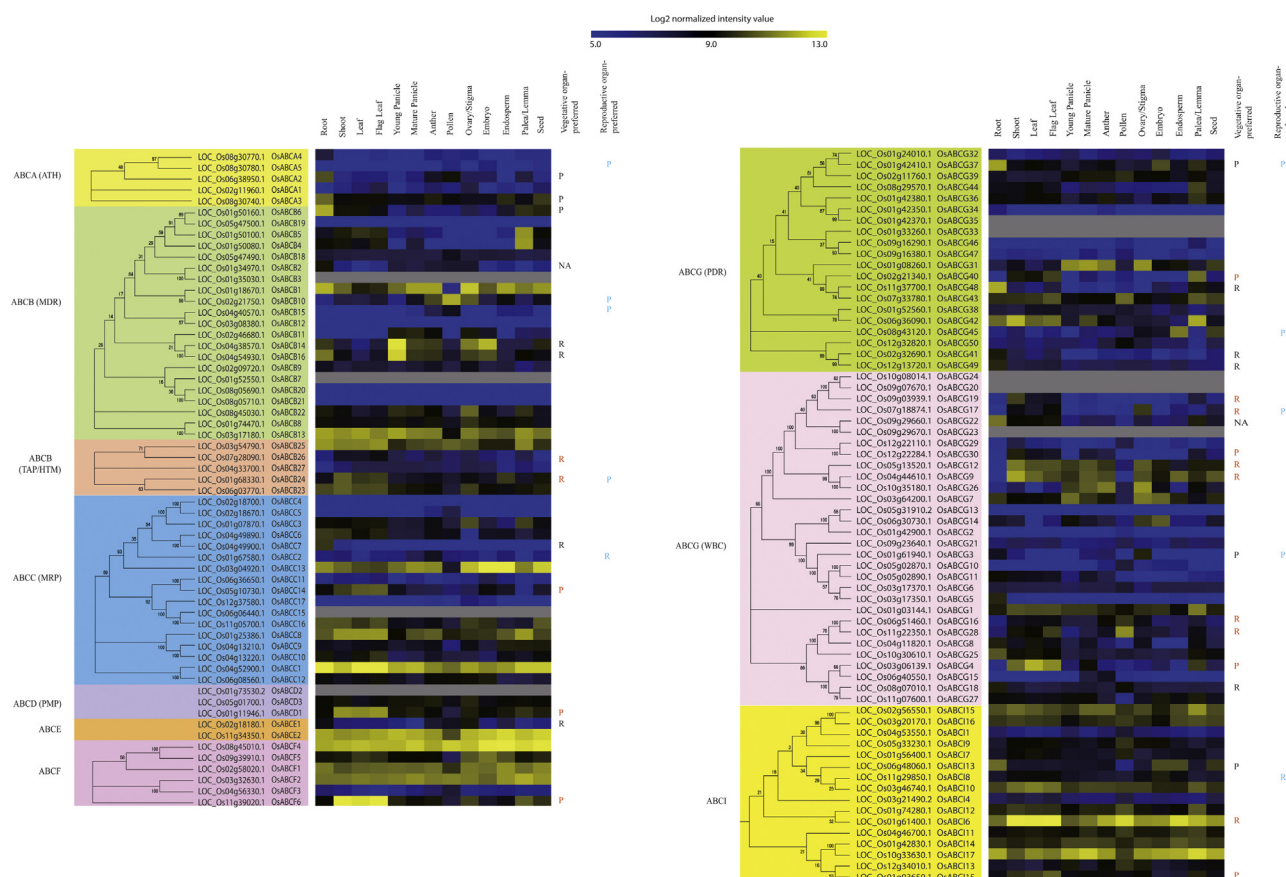


Fig. 2. Anatomic expression of rice ABC genes by meta-analysis of tissue specific expression profiles. The Affymetrix meta-analysis of tissue specific expression database based on 983 array data was divided into 13 anatomical tissues/organs. Blue denotes a low level of expression and yellow denotes a high level of expression. Gray denotes data not analyzed by microarray analysis. Predominant genes are indicated as P; non-predominant or redundant genes are indicated as R; and NA indicates not analyzed data due to the absence of the probes on Affymetrix array. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

for 1 min 30 s for 22–38 cycles. For real-time PCR, we used cycling conditions of 95 °C for 15 s, 57 °C for 30 s and 72 °C for 60 s. To validate the significant expression changes of genes in response to drought stress, we carried out real-time PCR analysis with three independent biological replicates. Relative transcript levels and fold change were calculated by $2^{-\Delta Ct}$ and $2^{-\Delta\Delta Ct}$ method (Schmittgen and Livak, 2008), respectively. Student t-test indicates the significant change under less than 0.05 and 0.01 *P*-values as asterisk (*) and double asterisk (**), respectively, in Figs. 7 and 8.

In silico network analysis

A hypothetical functional gene network mediated by rice ABC transporters was developed from RiceNet (<http://www.functionalnet.org/ricenet/>) (Lee et al., 2011) by querying the MSU locus IDs of 14 ABC transporter genes with confirmed drought response in Fig. 7. We uploaded the sif file from RiceNet to Cytoscape version 2.8.1 software. For whole genes in network, we also generated $\log_2$ fold-change data in response to abiotic stresses. We then integrated $\log_2$ fold-change data in response to abiotic stresses into nodes by using “File > Import > Attribute from Table” function. We assigned node color with criteria >1.5 and <-1.5 for up-regulation and down-regulation, respectively (COV < 1): up-regulated genes were marked as red color and down-regulated ones, green (Fig. 9).

Results

Phylogenetic analysis of ABC transporter family proteins in rice and *Arabidopsis thaliana*

Although several studies carried out genome-wide identification and analysis of rice ABC transporter genes, the number of rice ABC transporters with known biological functions is very limited. To get functional clues of rice ABC transporter family members, we tried to identify rice homologs of *Arabidopsis* ABC transporter genes with known function from previous studies. Recently, Verrier et al. (2008) updated the annotation of rice and *Arabidopsis* ABC transporters and identified 133 rice and 131 *Arabidopsis* ABC transporters. However, we did not find any common sites to compute a phylogenetic tree for 8 rice and 3 *Arabidopsis* ABC transporters through MEGA analysis. After eliminating these transporters which cause errors in developing comparative phylogenetic tree between rice and *Arabidopsis* ABC transporters, we produced a phylogenetic tree consisting of 125 rice and 128 *Arabidopsis* ABC transporters (Fig. 1). We then generated 8 subfamilies of rice and *Arabidopsis* ABC transporters as described in a previous study except ABCH which is absent from plant: ABCE, ABCB, ABCC, ABCD, ABCE, ABCF, ABCG, and ABCI. Of these, ABCB is further divided into ABCB-MDR and ABCB-TAP/ATM, and ABCG into ABCG-WBC and ABCG-PDR (Fig. 1). Interestingly, members of the ABCI subfamily were not clustered together but were distributed to clades closely linked to ABCE, ABCB-TAP/ATM, ABCG-PDR, and ABCG-WBC. This result indicates

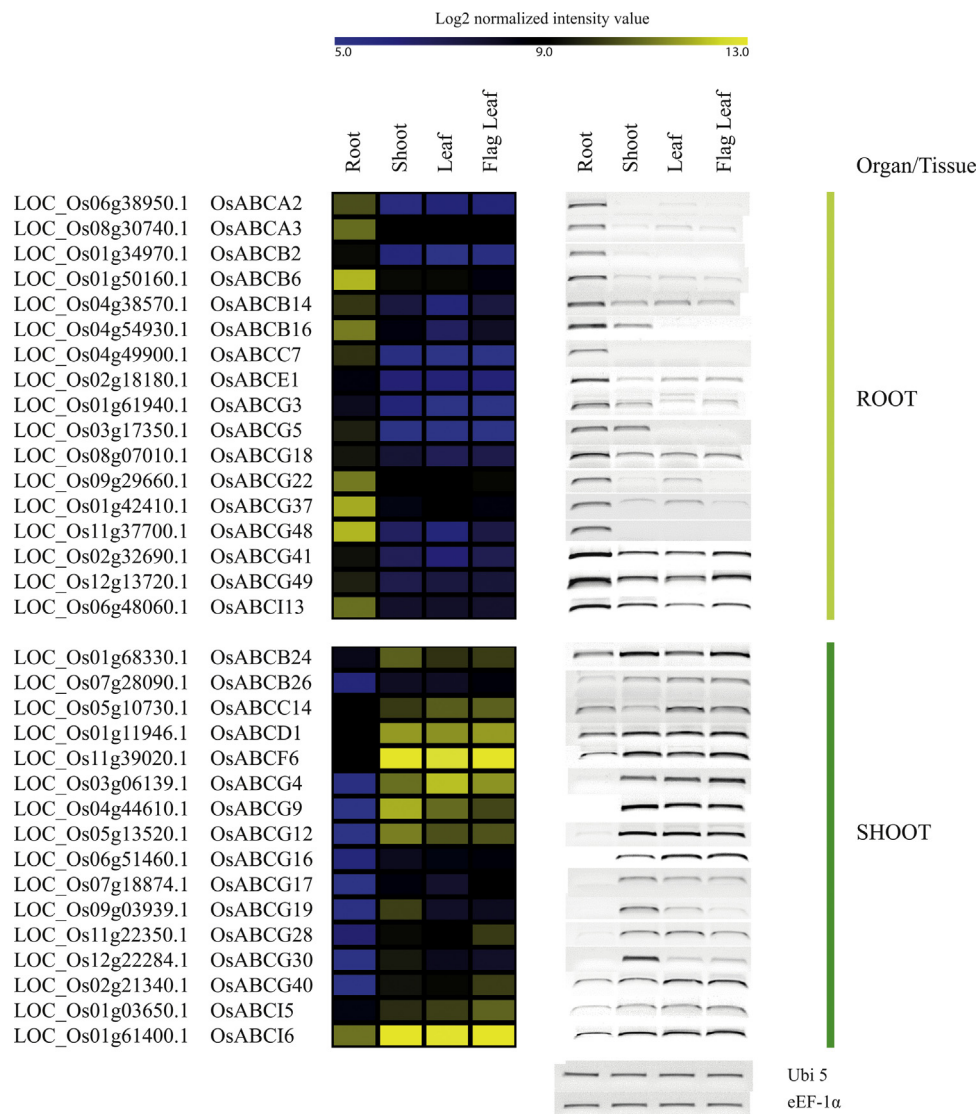


Fig. 3. Meta-analysis of tissue specific expression profiles of genes that show preferential expression in vegetative growth stage and RT-PCR analysis of selected genes. Organ/tissue denotes the type of tissue showing preferred expression during vegetative growth stage. Primers for RT-PCR are summarized in Table S4. *Rice Ubiquitin 5* (*OsUbi5*) and *Rice Elongation Factor-1α* (*OseEF-1α*) are internal controls for RT-PCR.

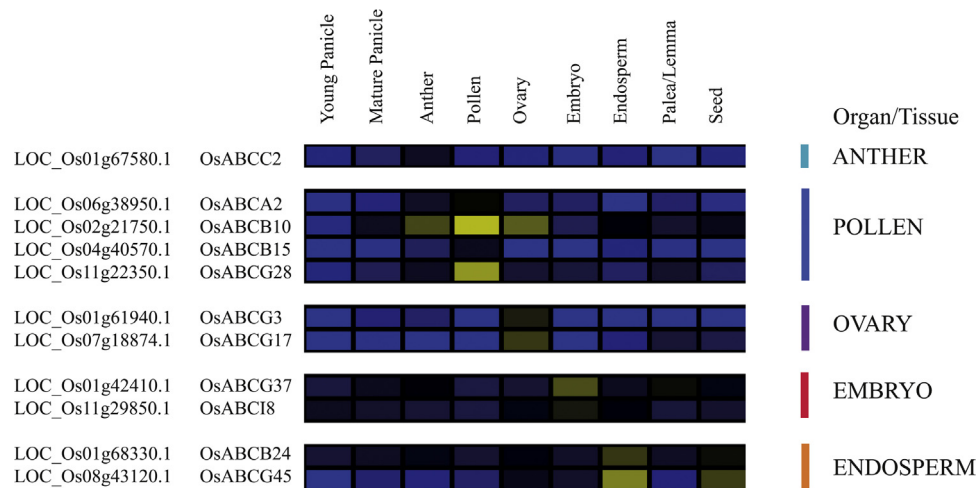


Fig. 4. Meta-expression analyses of genes that show preferential expression in reproductive growth stage.

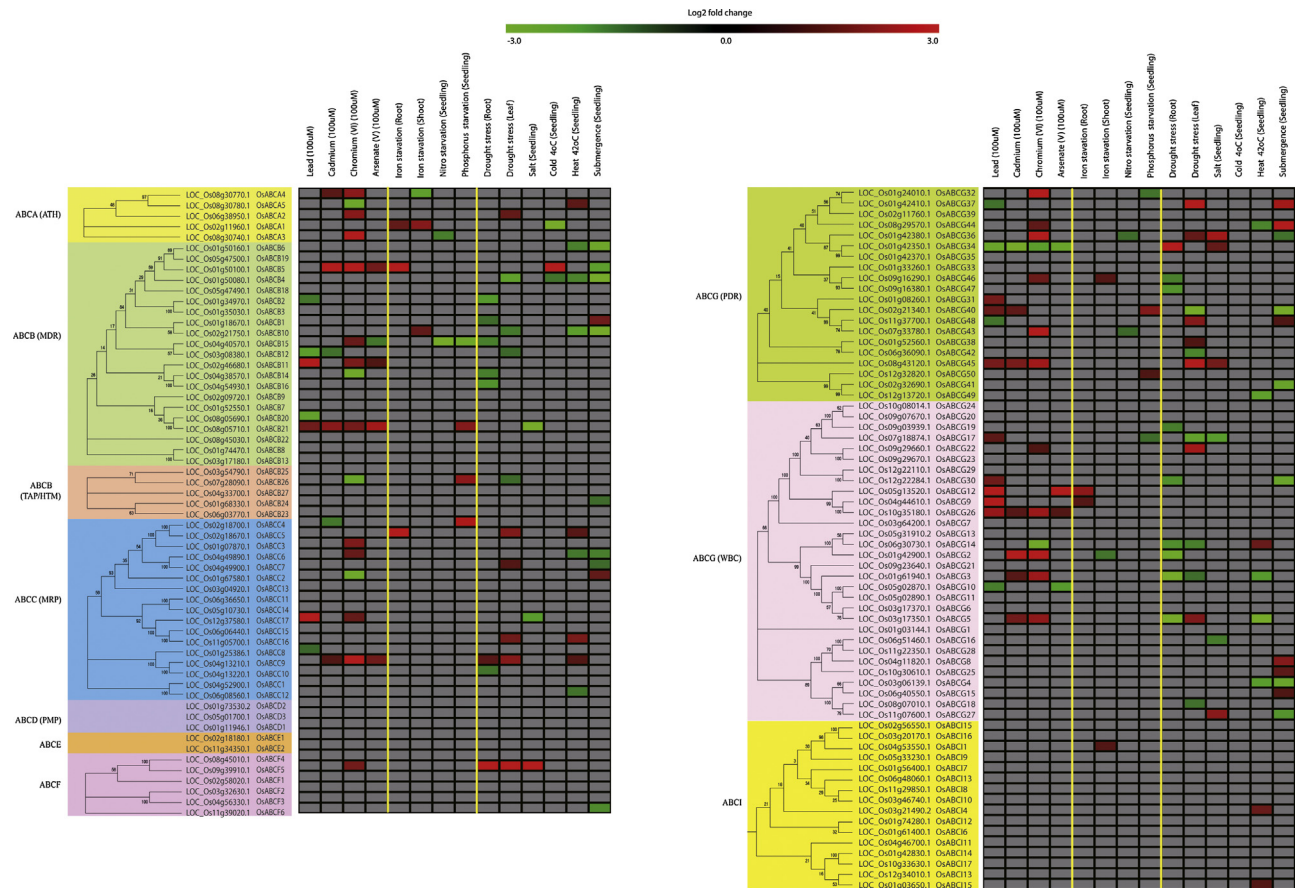


Fig. 5. Expression pattern of rice ABC transporter genes in response to stress conditions using Affymetrix arrays. Phylogenomic analysis of 125 rice ABC transporter genes with the integration of abiotic stress meta-analysis of tissue specific expression profiles data. The heat maps are divided into heavy metal stress, nutrient deficiency, and abiotic stress by yellow lines. Gray boxes denote a less significant fold-change or data not analyzed by microarray analysis. Red colored boxes indicate the up-regulation of applied stress treatment compared to control and green ones, down-regulation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

that this subfamily comprises various genes that encode one single domain such as NBD, or accessory domains, as previously described (Davidson et al., 2008). Except for the large divergence of ABCI, members of all other subfamilies were clustered together. Most of the rice ABC genes have at least one Arabidopsis ABC transporter in the same clade, suggesting possible conservation of function between monocot and dicot species. However, a few branches contain only rice ABC transporters without Arabidopsis homologs. For example, the cluster of ABCG-WBC consists of eight members from rice (*OsABCG17*, *OsABCG19*, *OsABCG20*, *OsABCG22*, *OsABCG23*, *OsABCG24*, *OsABCG29*, and *OsABCG30*) and that of ABCB-MDR consists of seven rice members (*OsABCB2*, *OsABCB3*, *OsABCB4*, *OsABCB5*, *OsABCB6*, *OsABCB18*, and *OsABCB19*). ABC transporter genes in these clusters might perform specified functions to rice and be potential candidate genes to distinguish monocot and dicot species.

Identification of tissue- or organ-preferred ABC transporter genes using meta-analysis of tissue specific expression profiles

The meta-analysis of tissue specific expression profiles were integrated into the phylogenetic tree context of eight subfamilies (Figs. S1 and 2). Tissue- or organ-preferred genes were identified by relative fold-change values in the selected tissue/organ compared with the other organs. As a result, during vegetative growth stage we identified 17 genes showing root-preferred expression and 16 genes showing leaf/shoot-preferred expression (Fig. 3). Of the former, we identified four ABCG-WBCs (*OsABCG3*, *OsABCG5*, *OsABCG18*, and *OsABCG22*), four ABCG-PDRs (*OsABCG37*,

OsABCG41, *OsABCG48*, and *OsABCG49*), four ABCBs (*OsABCB2*, *OsABCB6*, *OsABCB14*, and *OsABCB16*), two ABCAs (*OsABCA2* and *OsABCA3*), one ABCC (*OsABCC7*), one ABCE (*OsABCE1*), and one ABCI (*OsABCI13*). Of the latter, we identified eight ABCG-WBCs (*OsABCG4*, *OsABCG9*, *OsABCG12*, *OsABCG16*, *OsABCG17*, *OsABCG19*, *OsABCG28*, and *OsABCG30*), one ABCG-PDR (*OsABCG40*), two ABCBs (*OsABCB24* and *OsABCB26*), two ABCIs (*OsABCI5* and *OsABCI6*), one ABCD (*OsABCD1*), one ABCC (*OsABCI4*), and one ABCF (*OsABCF6*). Among the genes with tissue- or organ-preferred expression in the vegetative growth stage, 14 genes showed predominant expression over other closely related family members, 17 genes had other family members with similar expression level or predominant expression, and predominance of the expression of 2 genes was not estimated due to lack of expression data for closely linked ABC transporters on the Affymetrix array platform (Fig. 2).

To confirm the results of meta-analysis of tissue specific expression profiles, we selected 17 ABC transporter genes that showed root-preferred expression and 16 ABC transporter genes showed leaf/shoot-preferred expression during seedling development and performed RT-PCR (data shown in Fig. 3). These data agreed well with meta-analysis of tissue specific expression profiles based on a large collection of Affymetrix microarray data except for *OsABCG5*, which had an intermediate level of expression (Fig. 3). Therefore 22 of 23 genes showed consistency between RT-PCR and microarray data. This result indicates the high fidelity of meta-expression profiles based on a large collection of transcriptome data.

During the reproductive stage, we identified four pollen-preferred genes (*OsABCA2*, *OsABCB10*, *OsABCB15* and *OsABCG28*),

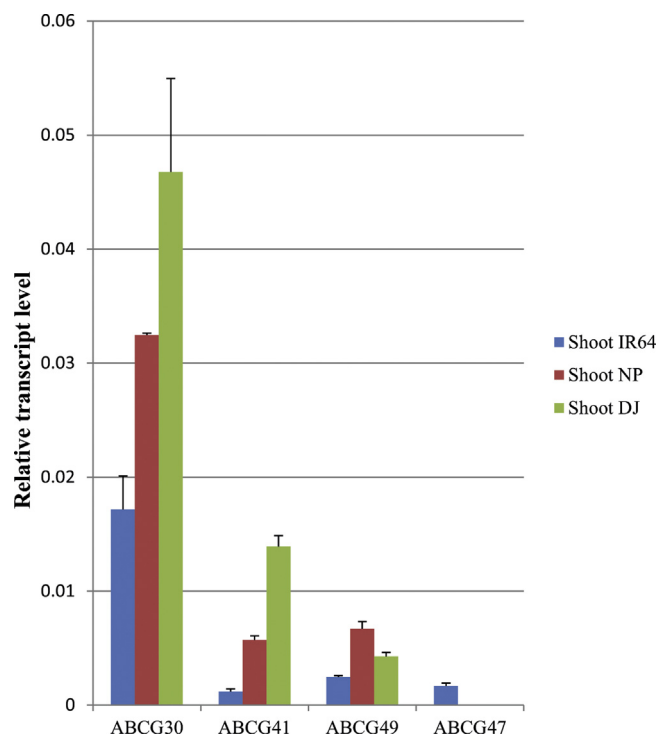


Fig. 6. Identification of *japonica*- or *indica*-preferred ABC transporters by real-time PCR analysis. *OsABCG47* showed leaf/shoot-preferred expression in *indica* samples, whereas *OsABCG30*, *OsABCG41*, and *OsABCG49* showed leaf/shoot-preferred expression in *japonica* samples. IR64: Indica-type rice variety IR64, NP: Japonica-type rice variety Nipponbare, DJ: Japonica-type rice variety Dongjin.

two ovary-preferred genes (*OsABCG3* and *OsABCG17*), two embryo-preferred genes (*OsABCG37* and *OsABC18*), two endosperm-preferred genes (*OsABCB24* and *OsABCG45*) and one anther-preferred gene (*OsABCC2*) ABC transporter gene (Fig. 4). Among the genes that showed tissue- or organ-preferred expression in the reproductive growth stage, eight genes showed predominant expression among other closely related family members, while three genes had family members with similar expression level or predominant expression (Fig. 2). The predominant members would be good targets for loss-of-function studies, whereas non-predominant members will require another strategy to overcome the functional redundancy among closely related gene family members, such as generating gain-of-function mutants, multiple gene silencing using RNAi, antisense, and artificial microRNA, and reporter gene assays.

Japonica- or *indica*-preferred ABC transporters

We identified three genes (*OsABCG30*, *OsABCG41*, and *OsABCG49*) that showed leaf/shoot-preferred expression in *japonica* samples, and one gene, *OsABCG47*, that showed leaf/shoot-preferred expression in *indica* samples. We measured the expression of these four genes by real-time PCR and confirmed the microarray data in Fig. S2 (Fig. 6). Therefore, we suggest the potential candidate genes of ABC transporters which might be useful to explain evolutionary difference between *japonica* and *indica* cultivars.

To explain the differential expression patterns of four genes (*OsABCG30*, *OsABCG41*, *OsABCG47*, and *OsABCG49*) between *japonica* and *indica* samples, we compared promoter regions between *japonica* and *indica* genomes for these genes. We extracted 2kb promoter regions of these genes from both *japonica* and *indica* genomes and compared them using Plantpan (Chang et al., 2008) (For details, see Materials and Methods). As a result, we found

that three (*OsABCG30*, *OsABCG41*, and *OsABCG49*) of them retain the difference of their promoter sequences between *japonica* and *indica* genomes for at least 10% of compared regions (Fig. S3). These differences might explain the differential expression patterns between *japonica* and *indica* samples. However, we could not find any difference in the promoter regions between *japonica* and *indica* genomes for *OsABCG47* except SNP at 1697 bp from the start codon and this variation might cause the disagreement of CARE-OSREP1 cis-element between *japonica* and *indica*: *indica* promoter of *OsABCG47* contains two adjacent cis-elements of CAREOSREP1 at 300 bp and 314 bp from 5' terminal, while the *japonica* promoter retains only one CAREOSREP1 cis-element at 314 bp (Fig. S3). Besides this change, the difference of epigenetic regulation between the *japonica* and *indica* promoter regions might cause the differential expression patterns between two cultivars. Further analysis remains to clarify our estimation based on comparative analysis of the promoter sequences.

Identification of ABC transporter genes that are differentially expressed under abiotic stresses or nutrient deficiency

We also investigated expression patterns of the rice ABC transporter genes in response to various stresses by integrating the fold-change data under stress conditions into the phylogenetic tree (Fig. 5). From this analysis, we predict that many of the rice ABC transporter genes have roles in responses to heavy metals, nutrient deficiency, and abiotic stresses. In total, we identified 73 genes that were differentially expressed under more than one stress treatment (Table S2). Log₂ fold-change data for severe heavy metal treatments such as 100 μ M Lead (Pb), Cadmium (Cd), Chromium (Cr), or Arsenate (As) were generated and analyzed (Dubey et al., 2010). We counted genes showing a log₂ fold-change higher than 1.5 or lower than -1.5 with less than 1 coefficient of variation (COV) as differentially expressed genes. Under a high concentration of Pb, 11 genes were up-regulated and another 8 showed down-regulation (Table S2). These genes belong to the ABCB, ABCC, and ABCG subfamilies. In a high concentration of Cd, genes encoding 10 ABC family members including 1 ABCA, 2 ABCBs, 1 ABCC, and 6 ABCGs were up-regulated (Table S2), and three (*OsABCB12*, *OsABCC4*, and *OsABCG34*) were down-regulated. Under a high concentration of Cr, 23 family members including 3 ABCAs, 4 ABCBs, 4 ABCCs, 1 ABCF, and 11 ABCGs showed up-regulation, while 6 family members including 1 ABCA, 2 ABCBs, 1 ABCC, and 2 ABCGs were down-regulated (Table S2). Six members including 3 ABCBs, 1 ABCC, and 2 ABCGs were up-regulated by high concentration of As (Table S2), and three (*OsABCB15*, *OsABCG10*, and *OsABCG34*) were down-regulated.

For nutrient starvation, we analyzed the log₂ fold-change data under conditions of starvation for iron (Fe), nitrogen (N), or phosphorus (P). Under Fe starvation, five genes (*OsABCA1*, *OsABCB5*, *OsABCC5*, *OsABCG9*, and *OsABCG12*) were up-regulated in root, and four genes (*OsABCA1*, *OsABCB10*, *OsABCG46*, and *OsABC11*) were up-regulated in shoot, whereas *OsABCA4* and *OsABCG2* were suppressed in shoot. Under N starvation, four genes (*OsABCA3*, *OsABCB15*, *OsABCG36*, and *OsABCG43*) were down-regulated with no members showing up-regulation. Under P starvation, five genes (*OsABCB21*, *OsABCB26*, *OsABCC4*, *OsABCG40*, and *OsABCG50*) showed up-regulation and three genes (*OsABCB15*, *OsABCG17*, and *OsABCG32*) were suppressed.

For abiotic stress responses, we analyzed the log₂ fold-change data under salt, drought, cold, and submergence stresses. In response to drought, 3 genes (*OsABCC9*, *OsABCF5*, and *OsABCG34*) are significantly up-regulated in root and 13 genes were up-regulated in leaf, while 14 genes in root and 7 in leaf were down-regulated. Of these, *OsABCG34* showed the highest up-regulation in root, followed by *OsABCF5*. Under salt stress, five genes (*OsABCF5*, *OsABCG27*, *OsABCG34*, *OsABCG36*, and *OsABCG45*) were

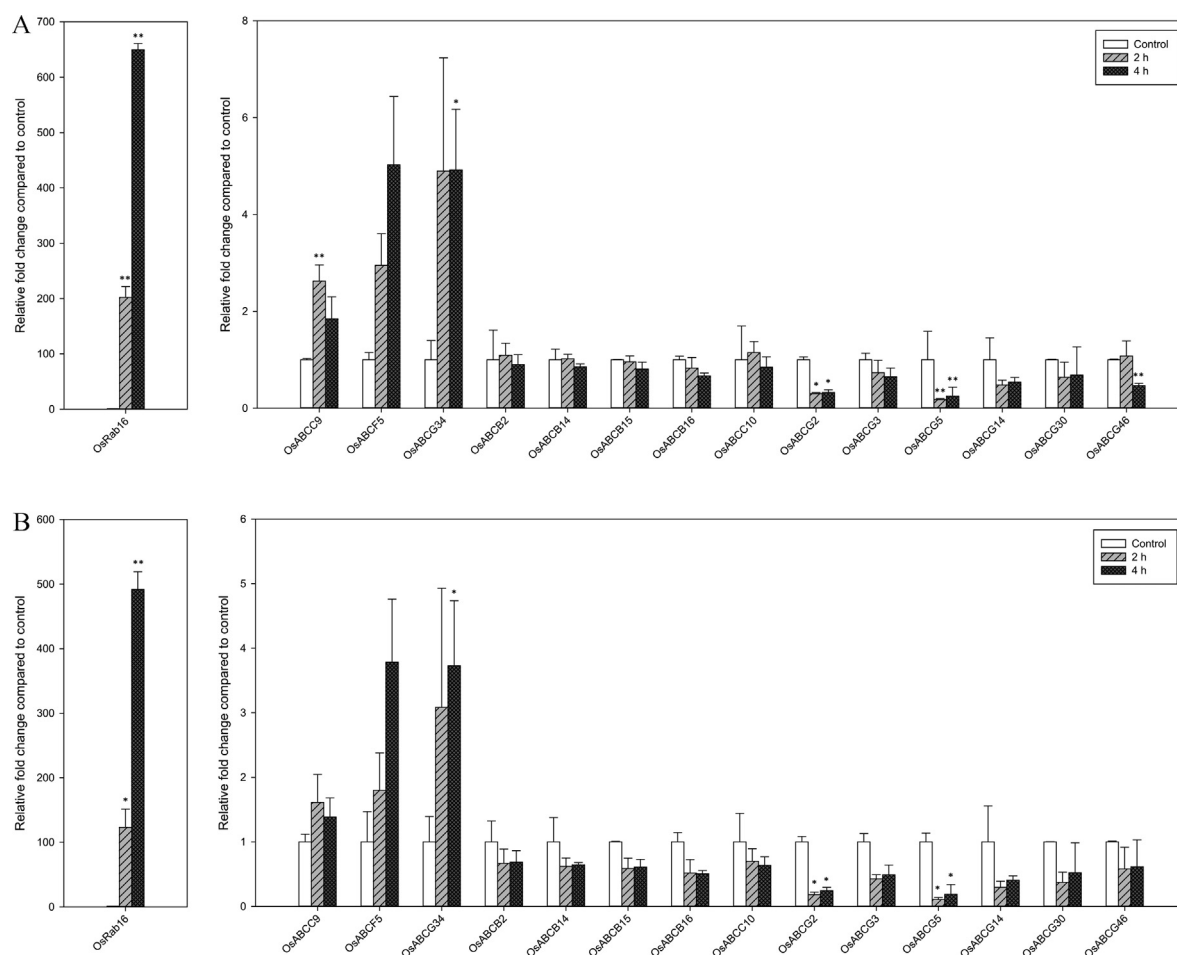


Fig. 7. Confirmation of significant changes in the expression of ABC transporters in root under drought stress by real-time PCR. Expression analysis of 14 drought-responsive ABC transporter genes was performed by real-time PCR. *OsRab16* is a positive control for drought stress. Expression levels of control were normalized as 1. (A) Relative fold change compared to control, *OsUbi5* was used as a reference gene. (B) Relative fold change compared to control, *OseEF-1α* was used as a reference gene. Asterisk (*) indicates P -value < 0.05; double asterisk (**), P -value < 0.01.

up-regulated, and four genes (*OsABCB21*, *OsABCC17*, *OsABCG16*, and *OsABCG17*) were down-regulated. *OsABCB5* was up-regulated under cold conditions, while *OsABCA1* and *OsABCB4* were down-regulated. Under heat stress, 7 genes showed up-regulation and 10 genes were down-regulated (Table S2). Under submergence, 8 genes were up-regulated and 14 genes were down-regulated (Table S2).

Of these, the genes responsive to heat stress belong to ABCB, ABCC, and ABCG subfamilies, and genes responsive to submergence stress belong to ABCB, ABCC, ABCF, and ABCG subfamilies. To summarize these data we generated a heat map showing the relationship between the type of stress and frequency of altered gene expression for each subfamily (Fig. S4). This showed that ABCC and ABCG-PDR subfamilies were more frequently up-regulated in response to drought in leaf, indicating that these proteins might be positively associated with drought stress in leaf. Conversely, ABCB-MDR and ABCG-PDR were more frequently down-regulated in response to drought in root, indicating a negative association with drought stress in root. ABCB-MDR was also frequently down-regulated in response to submergence. ABCB-MDR, ABCC, ABCG-WBC, and ABCG-PDR subfamilies were frequently up-regulated by high concentration of Cr. ABCG-WBC was also frequently up-regulated by high concentrations of Pb and Cd, indicating that this subfamily might play roles in transporting a diverse range of heavy metals.

Supplementary Fig. S4 related to this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jplph.2014.05.006>.

Validation of significant expression changes in rice ABC transporters in response to drought stress using real-time PCR

Among the many stresses that influence expression of ABC transporter genes, we are most interested in the response to drought stress. To validate the differential expression patterns observed in response to drought, we performed quantitative real-time PCR (q-PCR) for genes that were significantly differentially regulated by drought stress in root. First of all, we used *OsRab16* (*LOC.Os11g26790*) as a positive control for drought stress in root (Mundy and Chua 1988). As we expected, expression patterns of *OsRab16* was obviously induced by 2 h and 4 h drought treatment in root, indicating that our drought treatments were well evaluated by this marker gene. By qRT-PCR, we then observed up-regulation of 3 members (*OsABCC9*, *OsABCF5* and *OsABCG34*) and down-regulation of 11 members (*OsABCB2*, *OsABCB14*, *OsABCB15*, *OsABCB16*, *OsABCC10*, *OsABCG2*, *OsABCG3*, *OsABCG5*, *OsABCG14*, *OsABCG30*, and *OsABCG46*) in at least one of drought treatments in root. Of these, the differential expression patterns of *OsABCF5*, *OsABCG2*, and *OsABCG5* are more obvious than others. On the other hand, three (*OsABCB1*, *OsABCG19*, and *OsABCG47*) down-regulated by drought stress in root were not confirmed by real-time PCR analysis (data not shown). This result might be explained by the difference of genetic background, growth stage, and detailed stress treatment of our materials used for drought stress response compared with those in microarray data. We used Dongjin variety (*japonica* ssp.) and two data sets used for microarray data were

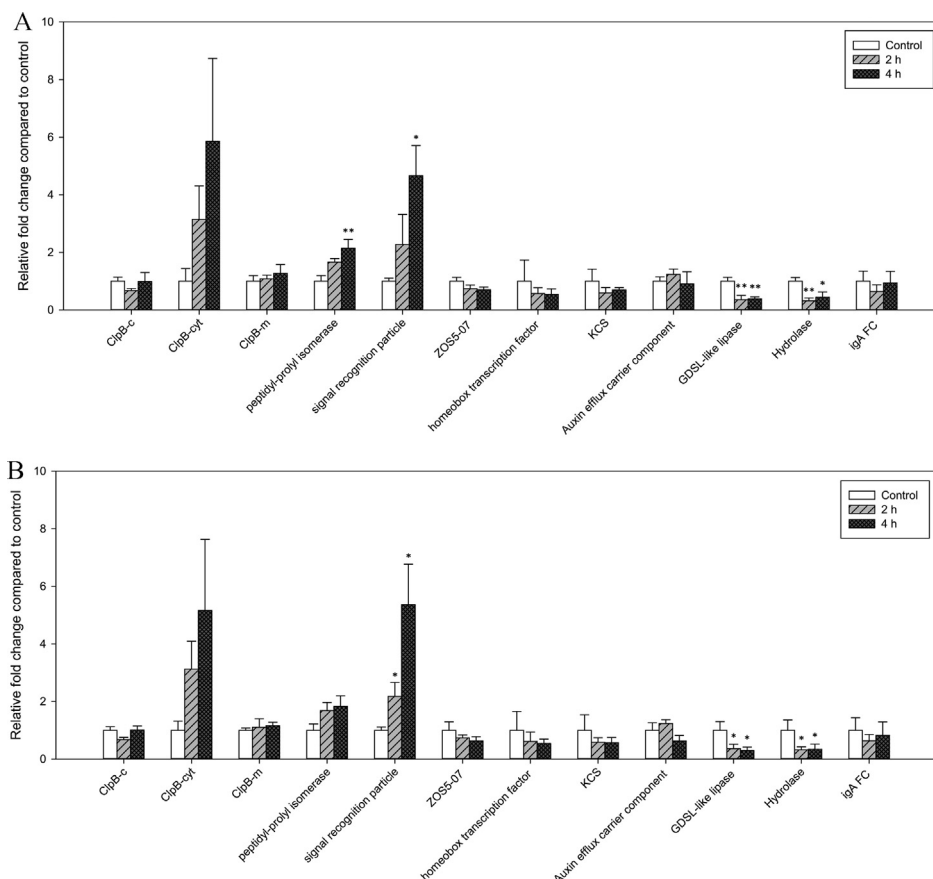


Fig. 8. Expression pattern analyses of 12 genes integrated in the network by real-time PCR. Expression patterns of 12 genes which associated with *OsABCF5*, *OsABCB14* and *OsABCG5* in the network including *ClpB-c*, *ClpB-cyt*, *ClpB-m*, peptidyl-prolyl isomerase, signal recognition particle, ZOS5-07, homeobox transcription factor, KCS, auxin efflux carrier component, GDSL-like lipase, hydrolase and iga FC were analyzed in root under drought treatment. Expression levels of control were normalized as 1. (A) Relative fold change compared to control, *OsUbi5* was used as a reference gene. (B) Relative fold change compared to control, *OseEF-1α* was used as a reference gene. Asterisk (*) indicates P -value < 0.05; double asterisk (**), P -value < 0.01.

produced using IR64, an *indica* variety (Table S3). This difference might explain inconsistent results between our q-RT-PCR data and referenced microarray data. In addition, the explanations include the difference of sampling time and growth stage of samples for the treatment. In total, we confirmed expression patterns of 14 genes as described in the microarray data (Fig. 7), indicating high fidelity of in silico expression analysis based on a large collection of qualified data sets.

Development of a functional gene network associated with ABC transporters that are differentially regulated under drought stress in root

We developed a functional network associated with 14 rice ABC transporter genes that are differentially expressed by drought stress in root using RiceNet database. The network consists of 268 linkages associated with 9 ABC transporter proteins (Fig. 9). To refine the network, the differential expression patterns in response to drought stress in root were integrated into the network (Fig. 9). There are two ABC transporters genes (*OsABCF5* and *OsABCG34*) were up-regulated and seven (*OsABCB14*, *OsABCB15*, *OsABCB16*, *OsABCG2*, *OsABCG3*, *OsABCG5*, and *OsABCG46*) were down-regulated. *OsABCF5* is associated with five up-regulated genes (*LOC_Os03g31300* encoding *ClpB-c*, *LOC_Os05g44340* encoding *clpB-cyt*, *LOC_Os02g08490* encoding *clpB-m*, *LOC_Os04g28420* encoding peptidyl-prolyl isomerase, and *LOC_Os05g43390* encoding signal recognition particle 54kDa protein). *OsABCB14* has 17 linkages that show down-regulation under drought in root.

Three of these, 3-ketoacyl-CoA (*LOC_Os03g14170*), oryzasin-1 (*LOC_Os01g44130*), and histone H3 (*LOC_Os05g36280*), shared linkages with other ABC transporters (*OsABCG5*; *OsABCB15*, and *OsABCG34*; and *OsABCF5*, respectively). *OsABCG5* has six linkages showing down-regulation in root under drought conditions, four of which showed common linkages with *OsABCB14* and *OsABCG3*. From the network, we also performed q-RT PCR for 5 genes which interact with *OsABCF5* and 4 genes interact with *OsABCB14*, and 4 genes interact with *OsABCG5* (Fig. 8). Of them, one gene is commonly linked with *OsABCB14* and *OsABCG5*. As a result, we found that all tested genes except *ClpB-c* showed similar trends with in silico expression data in root under drought stress. Of these, the differential expression patterns of *ClpB-cyt*, signal recognition particle protein, GDSL-like lipase, and hydrolase genes are much clearer than the others, suggesting the primary candidate genes for further functional analysis associated with ABC transporters in the network.

Discussion

Function of ABC transporters in rice based on meta-analysis of tissue specific expression profiles and functional homolog assignment

ABC transporters in plants constitute one of the largest protein families found in all living organisms (Henikoff, 1997). There are more than 125 ABC transporters in rice and the biological or molecular features of most of these have not yet been determined.

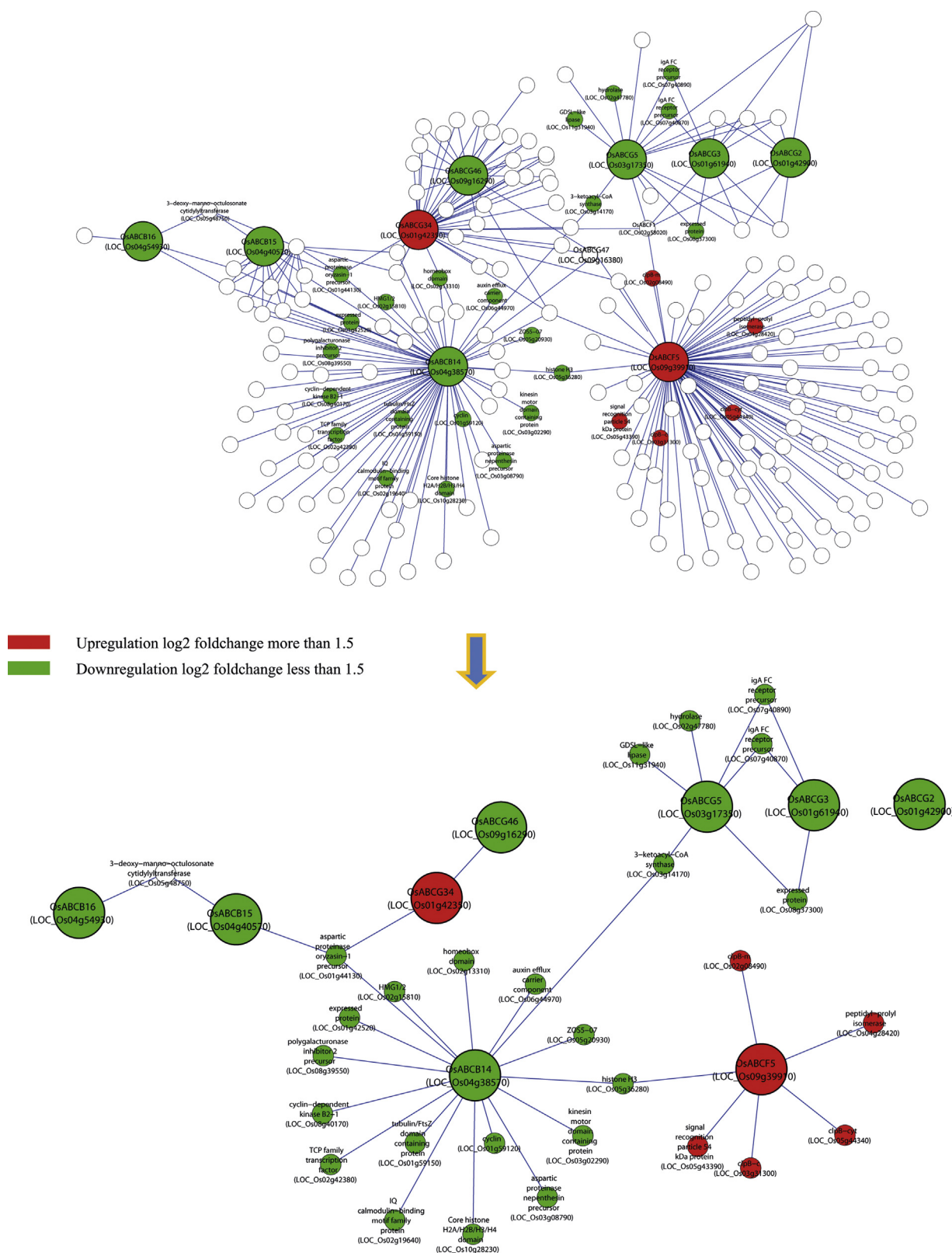


Fig. 9. Development and refinement of a functional gene network associated with ABC transporters that are responsive to drought in root. The functional gene network via nine ABC transporter proteins was developed: *OsABCF5* and *OsABCG34* showed up-regulation in response to drought in root; and *OsABCB14*, *OsABCB15*, *OsABCB16*, *OsABCG2*, *OsABCG3*, *OsABCG5*, and *OsABCG46* showed down-regulation. Red and green colors in nodes denote up-regulation with log₂ fold change ≥ 1.5 and down-regulation with log₂ fold-change ≤ -1.5 , respectively.

Even if comparative structural analyses of rice and Arabidopsis ABC transporters have been reported (Garcia et al., 2004; Jasinski et al., 2003; Verrier et al., 2008), genome-wide expression analysis for ABC transporter super-family in rice has not been carried out. Expression analysis for 25 ABCG subfamily members in response to abiotic stress and phytohormone treatments was only reported but this data cannot cover whole ABC super-family in rice (Matsuda et al., 2012). Therefore, our transcriptome data for tissue specific expression analyses and abiotic stress responses will be useful references for the further functional analyses. Hence, combination between phylogenetic analysis (homolog assignment) and microarray data does not only assist to detect gene expression of individual members but also support to more accurately predict the biological function. A typical example is *OsABCG15* that plays a critical role in exine formation and pollen development, and that *OsABCG15* expression was strongly detected in anther at the young microspore stage, weakly detected in the anther at the meiosis stage, and hardly detected in root, leaf, and the non-reproductive organs. Overall, these findings correspond to meta-analysis of tissue specific expression profiles based on Affymetrix data presented in this study (Fig. S2). Moreover, *OsABCG15* is an homologous gene of *AtABCG26* in Arabidopsis, which is involved in transferring sporopollenin lipid precursors from tapetal cells to anther locules, thus promoting exine formation on the pollen surface (Quilichini et al., 2010).

AtABCB19, the Arabidopsis homolog of *OsABCB14*, was characterized as having roles in auxin transportation in Arabidopsis seedlings present in the cortex and epidermis (Wu et al., 2010). *OsABCB14* itself shows strong expression in the root of vegetative stage plants, suggesting a similar role in auxin transportation in the root of rice. RiceXPro database serves the meta-expression analysis of rice roots and shoots in response to six phytohormones such as ABA, gibberellin, auxin, brassinosteroid, cytokinin, and jasmonic acid (Sato et al., 2013). As a result, we found that *OsABCB14* is slightly stimulated in roots and shoots by Auxin treatment but not by others. This result supports that *OsABCB14* is responsive to Auxin and probable transporter of auxin like Arabidopsis homolog, *AtABCB19*. In addition, our meta-analysis of tissue specific expression profiles indicated that *OsABCB10*, closely related to *OsABCB1*, is expressed at the highest level in anther and pollen, while *OsABCB1* was expressed strongly in almost all organs/tissues except pollen (Fig. S2). We expect that *OsABCB1* and *OsABCB10* perform molecular function but are modulated by different spatial and temporal regulation. Although further study is required to elucidate the function of *OsABCB10* in the reproductive organs, *OsABCB10* might play roles in development of mature anthers and pollens unlike *OsABCB1* with ubiquitous function in organs except pollens.

OsABCC1 and *OsABCC13* were ubiquitously expressed. *OsABCC1* shows a similar expression pattern to the Arabidopsis homolog, *AtABCC1/AtMRP1*, which functions in the transport of glutathione S-conjugate (Lu et al., 1997). Thus, *OsABCC1* might have similar function in rice with *AtABCC1/AtMRP1*. Expression of *OsABCC13/OsMRP5* was identified in a recent study and its mutation causes a low phytic acid phenotype in rice seeds (Xu et al., 2009). A single base pair change (C/G-T/A transition) in the sixth exon of this gene causes homozygous non-lethal phenotype but both a 5-bp deletion in the first exon and T-DNA insertion of this gene cause homozygous lethal phenotypes. We presume that severe defects of this gene with ubiquitous expression patterns are manifested first in developing seeds. Fine regulation of this gene under the control of diverse promoters that drive tissue-specific or stress-dependent expression might reveal the biological functions associated with the promoter activities besides seeds.

OsABCG9 and *OsABCG26* are closely clustered homologs of *AtABCG11/DESPERADO*, which is involved in the secretion of cutin and wax on the epidermis of leaf and is induced under salt, ABA,

and wound stress (Panikashvili et al., 2007). However, *OsABCG9* showed a high level of expression in all tissues except root and pollen, while *OsABCG26* was preferentially expressed in developing panicle, anther, and ovary/stigma (Figs. 2 and S2). These indicate that *OsABCG9* might play primary roles in the secretion of cutin and wax during both vegetative and reproductive stages, whereas *OsABCG26* might have a supplementary function during reproductive stages. On the other hand, the expression pattern of *OsABCG9* is similar to *OsABCG12*, which is the most homologous gene with *AtABCG12* in Arabidopsis. Panikashvili et al. (2007) found that *AtABCG12* is involved in the export of wax to the cuticle. Interestingly, the *atabcg11atabcg12* double knockout did not show a stronger phenotype than each of the single mutants, suggesting that the combination of two half-size ABCG transporters functions as a heterodimer. Together, these data suggest that *OsABCG9*, *OsABCG12*, and *OsABCG26* might play roles in the export of wax and cutin, like their homologs in Arabidopsis. Moreover, *OsABCG9* and *OsABCG12* might function in a heterodimeric form like *AtABCG11* and *AtABCG12* in Arabidopsis. Based on meta-analysis of tissue specific expression profiles and the closest linkage in phylogenetic tree, we expect that *OsABCG16* and *OsABCG28* have roles in mature anther or pollen development (Figs. 2 and S2).

Developmental or anatomical functions of other ABC transporters can be predicted based on integration of meta-analysis of tissue specific expression profiles into the phylogenetic tree context, which provides a more reliable estimation of the functional redundancy among these gene family members. Thus, a combination of the integrated phylogenetic tree with meta-analysis of tissue specific expression profiles and functional homolog assignments is an effective method to predict conserved biological function among multiple species. Further functional analysis will clarify the functional redundancy estimated by our phylogenomic analysis.

Roles of japonica- or indica-preferred ABC transporters

OsABCG47 showed leaf/shoot-preferred expression in *indica* samples, whereas *OsABCG30*, *OsABCG41*, and *OsABCG49* showed leaf/shoot-preferred expression in *japonica* samples (Figs. 6 and S2). These might be candidate genes to determine evolutionary differences between *japonica* and *indica*. For example, the *SNORKEL1* and *SNORKEL2* genes that encode ethylene-responsive transcription factors and the rice *PsbS* homologues 1 (*OsPsbS1*) gene are differentially expressed between *japonica* and *indica* (Hattori et al., 2009; Kasajima et al., 2011). Increased expression of *SNORKEL1* and *SNORKEL2* in *indica* cultivars stimulates shoot elongation under submergence (Hattori et al., 2009), whereas decreased expression of the *OsPsbS1* gene in *indica* cultivars leads to a defect in non-photochemical quenching capacity (Kasajima et al., 2011). Therefore, further analysis of cultivar-preferred ABC transporters will clarify their evolutionary roles during domestication.

Function of ABC transporters in rice under stress condition

In this study, fold-change data indicate that *OsABCF5* responds strongly under drought stress in root and leaf as well as during high salt stress in seedlings. In addition, *OsABCF5* also showed up-regulation under heavy metal Cr stress condition. The activation of *OsABCF5* under multi-stresses suggests that it might participate in many biological processes in order to rescue rice against diverse stress environments. However, because of the lack of a TMD it might function in processes other than transportation during dehydration or heavy metal stress.

OsABCG34 was down-regulated by metal heavy stresses including Pb, Cd, Cr, and Ar, but was significantly up-regulated in root by drought treatment. In addition, *OsABCG34* showed root-preferred expression in meta-analysis of tissue specific profiles. This suggests

that it functions in roots to process heavy metal stress or drought stress. Moreover, *OsABCG34* is the one of homologous genes of *AtABCG36/AtPDR8* which was identified as an efflux pump of Cd²⁺ or Cd conjugates at the plasma membrane (Kim et al., 2007). It is important for drought resistance (Kim et al., 2010), suggesting that *OsABCG34* might have conserved functions with *AtABCG36/AtPDR8*.

Functions of ABC transporters that are differentially regulated in root by drought stress using network analysis

The ABCF subfamily in plants has not been characterized functionally. Analysis using the functional gene network tool for rice study, RiceNet, revealed that *OsABCF5* is associated with three ClpBs/Hsp100 genes (*ClpB-c*, *ClpB-cyt*, *ClpB-m*) that are up-regulated in root by drought stress (Fig. 9). *ClpB-cyt* (LOC.Os05g44340) was previously reported to be induced by heat and drought (Hu et al., 2009). Wu et al. (2009) also noted that this ClpB promoter can be used to enhance drought and heat tolerance. Recently, all three ClpB genes were found to be induced in response to exogenous ABA application as well as oxidation stress (Singh et al., 2010). Peptidyl prolyl isomerase is considered as a molecular chaperone and showed the inducing expression by heat stress in bacteria and plants (Kurek et al., 1999; Nadeau et al., 1993). q-PCR result reveals all of them show up-regulated expression by drought stress in rice root (Fig. 8). Thus, further research is required to confirm the interaction or functional linkage between *OsABCF5* and molecular chaperones, although the data support the possibility that *OsABCF5* is associated with heat shock proteins as a potential modulator of responses to heat and drought.

OsABCB14 is predicted to be an auxin transporter and was down-regulated under drought in root. *ZOS5-07* (LOC.Os05g20930) belongs to the C₂H₂ zinc finger family of plant transcription factors. Some members of this family, such as ZPT2 related-genes (*AZF1*, *AZF2*, *AZF3* and *STZ*), are up-regulated by abiotic stress in Arabidopsis (Sakamoto et al., 2004). However, a previously unknown zinc finger transcription factor, Drought and Salt Tolerance (DST), negatively regulates stomatal closure via the regulation of genes involved in ROS homeostasis (Huang et al., 2009). Suppression of *ZOS5-07* under drought stress might enhance drought tolerance in rice through a pathway similar to that of the Arabidopsis homolog DST. Many of the Arabidopsis homeobox genes, including *ATHB6*, *ATHB7*, and *ATHB12*, have been shown to be induced in response to exogenous ABA and water deficiency (Himmelbach et al., 2002; Lee and Chun, 1998; Söderman et al., 1996). In particular, *ATHB6* has been identified as an interacting protein of protein phosphatase *ABI1* and acts as negative regulator of the ABA response (Himmelbach et al., 2002). Similarly, a homeobox transcription factor (*LOC.Os02g13310*) that is associated with *OsABCB14* in the network might be a potential candidate to improve the tolerance response under drought stress. Thus, it is possible that *OsABCB14* has important roles in auxin and ABA pathway.

OsABCB14 and *OsABCG5* have a common linkage, 3-ketoacyl-CoA synthase (*KCS*, LOC.Os03g14170). There are several studies indicating the involvement of 3-ketoacyl-CoA synthases such as *KCS20* and *KCS2/DAISY* in the drought stress response through up-regulated expression under drought and ABA treatment (Lee et al., 2009). Moreover, the mutant of *Wax Crystal-Sparse Leaf1*, which encodes a beta-ketoacyl CoA synthase, shows enhanced sensitivity to drought (Yu et al., 2008). Gigon et al. (2004) demonstrated that the reduction of lipid content in leaves was negatively affected by drought, while many ABCG proteins are involved in lipids transportation in Arabidopsis (Bird et al., 2007; Pighin et al., 2004; Quilichini et al., 2010). Thus, we expect that *OsABCB14* and *OsABCG5* might be required for the translocation of a precursor or itself of 3-ketoacyl-CoA. Interestingly, this study showed down-regulated expression of *OsABCG5* under drought, while Matsuda et al. (2012) reported

that *OsABCG5/Reduced culm number1* showed up-regulated expression following ABA treatment (Matsuda et al., 2012). This suggests that there might be an ABA-independent stress signaling pathway associated with *OsABCG5*. Because there are few studies or genome-wide data on ABC transporters and their functions under drought stress, it is difficult to predict their biological function. Further accumulation of genome-wide data and functional analysis for ABC transporter family in rice will elucidate their functions. Systematic functional studies using relating gene indexed mutants will elucidate the biological roles.

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